

AWS EC2, S3 and IAM Configuration

Introduction

The objective of this task was to practice some basic AWS services by launching an EC2 instance, creating an S3 bucket with a lifecycle policy, and creating a least-privilege IAM policy for a specific role.

We mainly focused on three important AWS services:

1. **Computing (EC2 instances, lambda)**
 2. **Storage (S3 buckets, EBS, EFS)**
 3. **Networking (VPC, Route S3)**
 4. **Identity and Access Management (IAM)**
 5. **Security (WAF, KMS)**
- **Region** – A geographical area where AWS has its data centers. Eg: Hyderabad.
 - **Availability Zone (AZ)** – A separate data center location within a Region. Eg: Northern , Southern areas of Hyderabad and so on.
1 Region can have 2+AZs.

1. Launch and Configure an EC2 Instance

Configuration

I launched an EC2 instance using the following configuration:

Setting	Configuration
Service	Amazon EC2
Instance Name	Hospitality-Server

Operating System	Amazon Linux
Instance Type	t2.micro
Network	Default VPC
Storage	8 GB
Security Group	Allow SSH (22)
Key Pair	Created/selected key pair

Steps

1. Open **Amazon EC2** in the AWS Management Console.
2. Select **Launch Instance**.
3. Enter the instance name.
4. Select **Amazon Linux** as the operating system.
5. Select the t2.micro instance type.
6. Select or create a key pair.
7. Configure the security group.
8. Configure the required storage.
9. Launch the instance.
10. Verify that the instance is in the **Running** state.

Result

The EC2 instance was successfully launched and configured. It can be used as a basic server for hosting or running an application.

2. Create an S3 Bucket with a Lifecycle Policy

Bucket Configuration

Setting	Configuration
Service	Amazon S3

Bucket Name	hospitality-training-bucket
Region	Selected AWS region
Versioning	Enabled
Public Access	Blocked

The bucket can be used to store files such as application data, reports, logs, or backups.

Lifecycle Policy

I created a lifecycle rule to automatically move older objects to a cheaper storage class.

Example rule:

- After **30 days** → Move objects to **S3 Standard-IA**
- After **90 days** → Move objects to **S3 Glacier Flexible Retrieval**

This helps reduce storage costs for data that is not accessed frequently.

Lifecycle Policy JSON

```
{
  "Rules": [
    {
      "ID": "ArchiveOldObjects",
      "Status": "Enabled",
      "Filter": {
        "Prefix": ""
      },
      "Transitions": [
        {
          "Days": 30,
          "StorageClass": "STANDARD_IA"
        }
      ]
    }
  ]
}
```

```

    },
    {
      "Days": 90,
      "StorageClass": "GLACIER"
    }
  ]
}
]
}

```

3. Least-Privilege IAM Policy

Objective

The purpose of the IAM policy is to give a role **only the permissions it needs** instead of giving it full access to AWS.

For this example, the role only needs to read objects from the specific S3 bucket.

IAM Policy JSON

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": "arn:aws:s3:::hospitality-training-
bucket/*"
    }
  ]
}

```

```
}  
]  
}
```

Explanation

- **Effect: Allow** – Allows the specified action.
- **Action: s3:GetObject** – Allows the role to read/download objects.
- **Resource** – Limits access to objects inside the specified S3 bucket.
- The role **cannot upload, delete, or modify** objects because those permissions were not granted.

This follows the **principle of least privilege**, where a user or role receives only the permissions required to perform its task.

COMPUTING:

What is an **EC2 instances**?

EC2 instances are virtual servers in AWS that we can create and use to run applications and other workloads in the cloud.

AMI (Amazon Machine Image) -> a template used to create EC2 instances.

EC2 Instance -> Virtual server created using that template

Auto-Scaling => increase or decrease the number of EC2 instances based on demand

STORAGE:

1. **Object Storage – S3**
2. **Block Storage – EBS**
3. **File Systems – EFS**

Object Storage (S3 Buckets): -> Stores objects/files

S3 buckets are containers in AWS used to store and organize files such as images, videos, documents, and backups in the cloud.

Block Storage (EBS): -> Provides disk storage for one EC2

EBS is a storage service in AWS that provides persistent storage for EC2 instances, similar to a hard disk attached to a computer.

File Systems (EFS): -> Provides shared file storage for multiple EC2s

EFS is an AWS file storage service that allows multiple EC2 instances to share and access the same files at the same time.

S3 Storage Classes:

1. **S3 Standard** – suitable for frequently accessed data.
2. **S3 Standard-IA** – suitable for data that is accessed less frequently.
3. **S3 Glacier** – suitable for long-term archival data.

S3 Lifecycle Policies -> helps in reducing storage costs and managing data efficiently

NETWORKING:

- VPC (Virtual Private Cloud) - Private AWS network
- Subnets - Smaller sections of a VPC
- Security Groups - Controls access
- Internet connectivity - Connects to the internet

IDENTITY AND SECURITY:

IAM – Identity and Access Management => used to control who can access AWS resources and what they are allowed to do. Basically acts like an identity of the user.

4. Overall Learning

Through this task, I learned how to launch and configure an EC2 instance, create an S3 bucket, and use lifecycle rules to manage stored data automatically. I also learned how IAM policies control access to AWS resources and how least-privilege permissions can improve security by giving a role only the access it actually needs.